

Optical Projection Tomography (OPT)

Cohn Group, GUDMAP Consortium

[3D anatomical reference models](#) of the genital tubercle, with or without gene-expression patterns for the marker gene Shh, were generated from TS19 to TS25 male and female embryonic mice using OPT.

Following microarray analysis of FACS purified Shh-Gfp mouse epithelial cells 86 urethral epithelium markers, restricted to the urethral palate, were identified. The 3D mouse reference models will be used to map the spatial domains of these gene expression patterns on to the reference genital tubercle models. This provides a 3D anatomical atlas of gene expression for the external genitalia and urethra allowing the gene expression patterns to be mapped and their distributions compared.

Preparation of Mouse Embryos

The protocol followed in the Cohn Lab is by James Sharpe published by Cold Spring Harbor Protocols (www.cshprotocols.org). To download the protocol for the preparation of mouse embryos for OPT imaging click here, <http://cshprotocols.cshlp.org/content/2011/6/pdb.prot5639.full>

OPT Imaging

Imaging of specimens was carried out using an OPT Scanner 3001M (Bioptronics, MRC Technology).